

FENT2321

Wind Energy and Design of a Wind Turbine

Assignment 7: Generator Design to match the blades - System design

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1 Problem 1: Operating point variables

1.1 Generator geometry

Poles	4
Phases	1
Coils	4
Length per coil	22.3m
Copper wire diameter	1.18mm
Number of turns per coil	50

1.2 Generator performance

1.2.1 Induced voltage at no load

RPM	E_A [V]
RPM _{design} - 300	73.3
RPM _{design}	83.4
RPM _{design} + 300	93.5